

Sequential Methods

Sequential experiments, introduced in Chapter 5, call for design decisions as data are collected. Optional stopping, in which the data are observed sequentially and used to decide when to terminate the experiment, would be the simplest example. A sequential approach can lead to increased efficiency, or it may achieve objectives not possible with a classical approach, but there are technical, practical, and philosophical issues that deserve attention.

Example 20.1. Sampling to a Foregone Conclusion. Let $X_1, X_2, \dots$ be i.i.d. from $N(\mu, 1)$, and let S_n denote the sum of the first n observations. The standard level α test of $H_0 : \mu = 0$ versus $H_1 : \mu \neq 0$ based on these observations will reject H_0 if $|S_n| > z_{\alpha/2}\sqrt{n}$.

Suppose a researcher proceeds sequentially, stopping the first time n that $|S_n|$ exceeds $z_{\alpha/2}\sqrt{n}$, so the sample size is

$$N = \inf\{n \geq 1 : |S_n| > z_{\alpha/2}\sqrt{n}\}.$$

Whenever N is finite, the classical test will reject H_0 . If $\mu \neq 0$, then N will be finite almost surely by the law of large numbers. In fact, N will also be finite almost surely if $\mu = 0$. To see this, note that for any k , $\{N = \infty\}$ implies

$$\{|S_k| \leq \sqrt{k}z_{\alpha/2}, |S_{2k}| \leq \sqrt{2k}z_{\alpha/2}\},$$

which in turn implies

$$\{|S_{2k} - S_k| \leq (\sqrt{k} + \sqrt{2k})z_{\alpha/2}\}.$$

These events have constant probability

$$p = P(|Z| \leq (1 + \sqrt{2})z_{\alpha/2}) < 1,$$

where $Z \sim N(0, 1)$, and so by independence,

$$P(N = \infty) \leq P\left(\bigcap_{j=1}^{\infty} \{|S_{2^{j+1}} - S_{2^j}| \leq (\sqrt{2^j} + \sqrt{2^{j+1}})z_{\alpha/2}\}\right) = \prod_{j=1}^{\infty} p = 0.$$

This example highlights one central technical problem with sequential experiments; sampling distributions may change with optional stopping. For any fixed sample size N , if $\mu = 0$, then $S_N/\sqrt{N} \sim N(0, 1)$, but with the random sample size N in the example, $|S_N/\sqrt{N}|$ exceeds $z_{\alpha/2}$ almost surely. Historically, there was controversy and concern when this was noted. If a researcher conducts an experiment sequentially, a standard frequentist analysis is not appropriate. For a proper frequentist analysis there must be a specific protocol detailing how the sample size will be determined from the data, so that the effects of optional stopping can be taken into account properly when probabilities, distributions, and moments of statistics are computed.

Surprisingly, likelihood functions after a sequential experiment are found in the usual way. Since Bayesian inference is driven by the likelihood, posterior distributions will be computed in the usual fashion, and a sequential design will not affect Bayesian analysis of the data. Due to this, design problems are often more tractable with a Bayesian formulation.

In Section 20.1 a central limit theorem is derived for sequential experiments and used to find stopping rules that allow asymptotic interval estimation with specified fixed accuracy. Section 20.2 studies stopping times in a more formal fashion, explaining why they do not affect likelihood functions. In Section 20.3, the backwards induction method, used to find optimal stopping times, is explored, focusing on a Bayesian approach to hypothesis testing. Section 20.4 introduces Wald's sequential probability ratio test for simple versus simple testing in a sequential context. Finally, Section 20.5 explores design issues beyond optional stopping, specifically stochastic approximation recursions in which independent regression variables are chosen adaptively, and "bandit" allocation problems.

20.1 Fixed Width Confidence Intervals

Let $X_1, X_2, \dots$ be i.i.d. from a one-parameter exponential family with marginal density

$$f_\theta(x) = h(x) \exp\{\eta(\theta)T(x) - B(\theta)\}.$$

Let $T_i = T(X_i)$ and $\bar{T}_n = (T_1 + \dots + T_n)/n$. Then the maximum likelihood estimator $\hat{\lambda}_n$ of a parameter $\lambda = g(\theta)$ based on the first n observations is a function of $\bar{T}_n$,

$$\hat{\lambda}_n = \hat{\lambda}(\bar{T}_n),$$

and by the delta method,

$$\sqrt{n}(\hat{\lambda}_n - \lambda) \Rightarrow N(0, \nu^2(\theta)),$$

where

$$\nu^2(\theta) = [\hat{\lambda}'(\mu_T(\theta))]^2 \sigma_T^2(\theta),$$

with $\mu_T(\theta) = E_\theta T_i$ and $\sigma_T^2(\theta) = \text{Var}_\theta(T_i)$. Using this, if $\nu(\cdot)$ is continuous and $\hat{\theta}_n$ is the maximum likelihood estimator of θ , then

$$\left(\hat{\lambda}_n \pm \frac{z_{\alpha/2}}{nu(\hat{\theta}_n)} / \sqrt{n} \right) \quad (20.1)$$

is an asymptotic $1 - \alpha$ confidence interval for λ .

If a researcher is interested in estimating λ with fixed precision, a confidence interval with a fixed width w would be desired. Since $\nu(\theta)$ will generally vary with θ , the interval (20.1) from any fixed sample may fail. But following a sequential strategy, the researcher may choose to continue sampling until the width $2z_{\alpha/2}\nu(\hat{\theta}_n)/\sqrt{n}$ of interval (20.1) is less than w . This leads to a sequential experiment with sample size

$$N = N_w = \inf \{ n : w^2 n \geq 4z_{\alpha/2}^2 \nu^2(\hat{\theta}_n) \}. \quad (20.2)$$

If w is small, N will be large, and it seems reasonable to hope that the interval

$$\left(\hat{\lambda}_N \pm \frac{z_{\alpha/2} \nu(\hat{\theta}_N)}{\sqrt{N}} \right) \quad (20.3)$$

from a sequential experiment will have coverage approximately $1 - \alpha$. And by construction, the width of this interval is at most w . This is correct, but a proper demonstration takes a bit of care, because the sample size N is a random variable, whereas sample sizes in our prior results on weak convergence were constant. The main result we need is a central limit theorem due to Anscombe (1952) in which the number of summands is random. Almost sure convergence, introduced in Section 8.7, and the strong law of large numbers play a role here.

The proposed sample size N_w in (20.2) tends to ∞ almost surely as $w \downarrow 0$. If $\hat{\theta}_n \rightarrow \theta$ almost surely,¹ then $\hat{\theta}_N \rightarrow \theta$ almost surely and $\hat{\theta}_{N-1} \rightarrow \theta$ almost surely. Since

$$4z_{\alpha/2}^2 \nu^2(\hat{\theta}_N) \leq w^2 N < w^2 + 4z_{\alpha/2}^2 \nu^2(\hat{\theta}_{N-1}),$$

it follows that

$$w^2 N \rightarrow 4z_{\alpha/2}^2 \nu^2(\theta)$$

almost surely as $w \downarrow 0$. If we define

$$n_w = \left\lfloor \frac{4z_{\alpha/2}^2 \nu^2(\theta)}{w^2} \right\rfloor,$$

then $w^2(N - n_w) \rightarrow 0$ almost surely as $w \downarrow 0$. The idea behind Anscombe's central limit theorem is that a shift in the sample size from N to n_w will change the limiting variable by an amount that is $o_p(1)$.

¹ When η is continuous, $\hat{\theta}_n$ is a continuous function of $\overline{T}_n$, and this follows from the strong law of large numbers.

Definition 20.2. *Random variables W_n , $n \geq 1$, are uniformly continuous in probability (u.c.i.p.) if for all $\epsilon > 0$ there exists $\delta > 0$ such that*

$$P\left\{\max_{0 \leq k \leq n\delta} |W_{n+k} - W_n| \geq \epsilon\right\} < \epsilon, \quad \text{for all } n \geq 1.$$

Theorem 20.3 (Anscombe). *If N_w , $w > 0$, are positive integer-valued random variables with $w^2 N_w \xrightarrow{P} c \in (0, \infty)$ as $w \downarrow 0$, if $n_w = \lfloor c/w^2 \rfloor$, and if W_n , $n \geq 1$ are u.c.i.p., then*

$$W_{N_w} - W_{n_w} \xrightarrow{P} 0$$

as $w \downarrow 0$.

Proof. Fix $\epsilon > 0$. For any $\delta > 0$,

$$P(|W_{N_w} - W_{n_w}| > \epsilon) \leq P(w^2 |N_w - n_w| > \delta) + P\left(\max_{w^2 |n - n_w| \leq \delta} |W_n - W_{n_w}| > \epsilon\right).$$

The first term here tends to zero regardless of the choice of δ . By the triangle inequality, if $m = \lceil n_w - \delta/w^2 \rceil$ (the smallest integer m with $w^2 |m - n_w| \leq \delta$), then

$$|W_n - W_{n_w}| \leq |W_n - W_m| + |W_{n_w} - W_m|,$$

and so

$$\max_{w^2 |n - n_w| \leq \delta} |W_n - W_{n_w}| \leq 2 \max_{w^2 |n - n_w| \leq \delta} |W_n - W_m|.$$

Therefore

$$P\left(\max_{w^2 |n - n_w| \leq \delta} |W_n - W_{n_w}| > \epsilon\right) \leq P\left(\max_{w^2 |n - n_w| \leq \delta} |W_n - W_m| > \epsilon/2\right).$$

Since the W_n are u.c.i.p., this probability will be less than $\epsilon/2$ if δ is sufficiently small, and the theorem follows as ϵ is arbitrary. $\square$

In Theorem 20.3, if $W_n \Rightarrow W$, $W_{n_w} \Rightarrow W$, and so

$$W_{N_w} = W_{n_w} + o_p(1) \Rightarrow W.$$

One example of particular interest would be normalized partial sums, $W_n = \sqrt{n} \bar{Y}_n$ with $Y_1, Y_2, \dots$ i.i.d. mean zero, and $\bar{Y}_n$ the average of the first n of these variables. The following maximal inequality, due to Kolmogorov, is used to show these variables are u.c.i.p. Let $S_k = Y_1 + \dots + Y_k$.

Lemma 20.4. *If $Y_1, \dots, Y_n$ are i.i.d. with mean zero and common variance $\sigma_Y^2 \in (0, \infty)$, then for any $c > 0$,*

$$P\left(\max_{1 \leq k \leq n} |S_k| \geq c\right) \leq \frac{n\sigma_Y^2}{c^2}.$$

Proof. Let A_k be the event that S_k is the first partial sum with magnitude at least c , that is,

$$A_k = \{|S_1| < c, \dots, |S_{k-1}| < c, |S_k| \geq c\}.$$

Because A_k is determined by $Y_1, \dots, Y_k$, $S_k 1_{A_k}$ is independent of $S_n - S_k = Y_{k+1} + \dots + Y_n$, and for $k \leq n$

$$E[S_k(S_n - S_k) 1_{A_k}] = E[S_k 1_{A_k}] \times E[S_n - S_k] = 0.$$

But on A_k , $S_k^2 \geq c^2$, and so for $k \leq n$,

$$E[S_n^2 1_{A_k}] = E[(S_n - S_k)^2 + 2S_k(S_n - S_k) + S_k^2] 1_{A_k} \geq c^2 P(A_k).$$

Since $\{\max_{1 \leq k \leq n} |S_k| \geq c\}$ is the disjoint union of $A_1, \dots, A_n$,

$$c^2 P(\max_{1 \leq k \leq n} |S_k| \geq c) = \sum_{k=1}^n c^2 P(A_k) \leq E\left[S_n^2 \sum_{i=1}^n 1_{A_i}\right] \leq ES_n^2 = n\sigma_Y^2,$$

proving the lemma. $\square$

Considering the normalized partial sums, since

$$|W_{n+k} - W_n| = \left| \frac{1}{\sqrt{n}} \sum_{i=n+1}^k Y_i - \left(\sqrt{\frac{n+k}{n}} - 1 \right) W_{n+k} \right|,$$

we have

$$\begin{aligned} P(\max_{0 \leq k \leq n\delta} |W_{n+k} - W_n| \geq \epsilon) &\leq P((\sqrt{1+\delta} - 1)|W_{n+k}| \geq \epsilon/2) \\ &\quad + P\left(\max_{0 \leq k \leq n\delta} \left| \sum_{i=n+1}^{n+k} Y_i \right| \geq \epsilon\sqrt{n}/2\right). \end{aligned}$$

By Chebyshev's inequality, the first term here is at most

$$\frac{4(\sqrt{1+\delta} - 1)^2 \sigma_Y^2}{\epsilon^2},$$

and by Lemma 20.4 the second term is at most

$$\frac{4\sigma_Y^2 \delta}{\epsilon^2}.$$

These bounds tend to zero as $\delta \downarrow 0$, uniformly in n , and so W_n , $n \geq 1$, are u.c.i.p.

Returning to fixed width interval estimation and the coverage probability for interval (20.3), by Theorem 20.3,

$$\sqrt{N}(\bar{T}_N - \mu(\theta)) \Rightarrow N(0, \text{Var}_\theta(T_i))$$

as $w \downarrow 0$, and by the delta method,

$$\sqrt{N_w}(\hat{\lambda}_N - \lambda) \Rightarrow N(0, \nu^2(\theta)).$$

Since $N \rightarrow \infty$ as $w \downarrow 0$ and $\hat{\theta}_n \rightarrow \theta$ as $n \rightarrow \infty$, both almost surely, $\hat{\theta}_N \rightarrow \theta$ almost surely as $w \downarrow 0$. It then follows that

$$\frac{\sqrt{N}(\hat{\lambda}_N - \lambda)}{\nu(\hat{\theta}_N)} \Rightarrow N(0, 1)$$

as $w \downarrow 0$. So the coverage probability for the confidence interval (20.3),

$$P_\theta \left[\lambda \in (\hat{\lambda}_N \pm z_{\alpha/2} \nu(\hat{\theta}_N) / \sqrt{N}) \right] = P_\theta \left(\frac{\sqrt{N}|\hat{\lambda}_N - \lambda|}{\nu(\hat{\theta}_N)} < z_{\alpha/2} \right),$$

converges to $1 - \alpha$ as $w \downarrow 0$.

20.2 Stopping Times and Likelihoods

In Chapter 5 we had some trouble representing data from a sequential experiment as a random vector, because this kind of experiment's sample size is not a fixed constant. The most elegant and standard way to ameliorate this problem is to use σ -fields to represent information. To understand how this is done, consider an experiment in which a coin is tossed two times, so the sample space is

$$\mathcal{E} = \{TT, TH, HT, HH\}.$$

Let $\mathcal{F}$ be the σ -field of all subsets of $\mathcal{E}$, $\mathcal{F} = 2^\mathcal{E}$, and let the random variable X give the number of heads. If we observe X we will know if certain events occur. For instance, we will know whether

$$\{X = 1\} = \{HT, TH\}$$

occurs. But other events, such as $\{HH, HT\}$ (the first toss lands heads), will remain in doubt. The collection of all events we can resolve,

$$\begin{aligned} \sigma(X) = \{ & \emptyset, \{TT\}, \{HT, TH\}, \{HH\}, \{TT, HT, TH\}, \\ & \{TT, HH\}, \{HT, TH, HH\}, \{TT, TH, HT, HH\} \}, \end{aligned}$$

is a σ -field. A means to learn which events in $\sigma(X)$ occur would provide exactly the same information about the outcome e as the value for X . Thus X and $\sigma(X)$ in a natural sense provide the same information.

The notions in the coin tossing example generalize easily. If we observe a random vector X , then we will know whether $\{X \in B\} = X^{-1}(B)$ occurs.

Here we insist that B is a Borel set to guarantee that $\{X \in B\}$ is an event in $\mathcal{F}$. We can then define

$$\sigma(X) \stackrel{\text{def}}{=} \{X^{-1}(B) : B \text{ Borel}\},$$

and it is easy to show that $\sigma(X)$ is a σ -field.

Consider now an experiment in which random vectors $X_1, X_2, \dots$ are observed sequentially. Let

$$\mathcal{F}_n = \sigma\{X_1, \dots, X_n\}, \quad (20.4)$$

the events we can resolve observing the first n variables, and take $\mathcal{F}_0 = \{\emptyset, \mathcal{E}\}$. These σ -fields are increasing, $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots$. In general, any increasing sequence of σ -fields $\mathcal{F}_n$, $n \geq 0$, is called a *filtration*. The filtration given by (20.4) would be called the *natural filtration* for X_i , $i \geq 1$. We can also define $\mathcal{F}_\infty$ as the smallest σ -field containing all events in $\bigcup_{n \geq 1} \mathcal{F}_n$. For the natural filtration, $\mathcal{F}_\infty$ would represent the information available from all the X_i . This σ -field may not equal the underlying σ -field $\mathcal{F}$; it will be strictly smaller if $\mathcal{F}$ contains events that cannot be determined from the X_i .

Sample sizes for a sequential experiment cannot depend on the observations in an arbitrary fashion. For instance, a design calling for two observations if and only if $X_5 > 10$ would be absurd. Clairvoyance needs to be prohibited in the mathematical formulation. In particular, the decision to stop or continue after n observations needs to be based on the information $\mathcal{F}_n$ available from those data. Specifically, the event $\{N = n\}$ should lie in $\mathcal{F}_n$. These variables are called *stopping times* according to the following definition.

Definition 20.5. A random variable² N taking values in $\{0, 1, 2, \dots, \infty\}$ is called a *stopping time with respect to a filtration* $\mathcal{F}_n$, $n \geq 0$, if

$$\{N = n\} \in \mathcal{F}_n, \quad \text{for all } n \geq 0.$$

Next we would like to find a σ -field that represents the information available when data are observed until a stopping time N . Any event $B \in \mathcal{F}$ can be written as the disjoint union of the sets $B \cap \{N = n\}$, $n = 1, 2, \dots, \infty$. If we can determine B from the data, it must be the case that that part of B where $N = n$ (i.e., $B \cap \{N = n\}$) must lie in $\mathcal{F}_n$, for any n , and we define

$$\mathcal{F}_N = \{B : B \cap \{N = n\} \in \mathcal{F}_n, \forall n = 0, 1, \dots, \infty\}.$$

It is not hard to show that $\mathcal{F}_N$ is a σ -field, and it represents the information available observing the data until stopping time N . We may also want to consider what random variables Y are based on the observed data. Because the event $\{Y \in B\} = Y^{-1}(B)$ can be resolved by observing Y , this event

² Since “ $+\infty$ ” is an allowed value for N , it may be slightly more proper to call N an extended random variable.

should lie in $\mathcal{F}_N$. But this requirement is simply that Y is $\mathcal{F}_N$ measurable. For instance, with the natural filtration, the stopping time N is $\mathcal{F}_N$ measurable, and $\bar{X}_N = (X_1 + \cdots + X_N)/N$ is $\mathcal{F}_N$ measurable. But X_{N+1} is not.

If we use σ -fields to represent information, we will also be interested in conditioning to revise probabilities and expectations in light of the information from a σ -field. With a random vector X and an integrable random variable Y , $E(Y|X)$ should be a measurable function of X , and smoothing must work for $YI\{X \in B\}$ with B an arbitrary Borel set:

$$EYI\{X \in B\} = EI\{X \in B\}E(Y|X),$$

for all Borel sets B . The requirements for conditioning on a σ -field $\mathcal{G}$ are similar. First, $E(Y|\mathcal{G})$ should be $\mathcal{G}$ measurable (based only on information available from $\mathcal{G}$); and second, smoothing should work for $Y1_B$ with B any event in $\mathcal{G}$:

$$EY1_B = E1_BE(Y|\mathcal{G}), \quad B \in \mathcal{G}.$$

The next result, Wald's fundamental identity, is the basis for likelihood calculations. In this result, there are two probability measures P_0 and P_1 . Let f_{0n} and f_{1n} denote joint densities for $(X_1, \dots, X_n)$ under P_0 and P_1 , and let $L_n = L_n(X_1, \dots, X_n)$ denote the likelihood ratio

$$L_n(X_1, \dots, X_n) = \frac{f_{1n}(X_1, \dots, X_n)}{f_{0n}(X_1, \dots, X_n)}.$$

Theorem 20.6 (Wald's Fundamental Identity). *If $f_{1n} = 0$ whenever $f_{0n} = 0$, and if $P_0(N < \infty) = P_1(N < \infty) = 1$, then*

$$P_1(B) = E_0 1_B L_N, \quad \forall B \in \mathcal{F}_N.$$

Proof. Because $\{N = n\} \cap B \in \mathcal{F}_n$, by Lemma 12.18,

$$P_1(N = n, B) = E_0 I\{N = n\} 1_B L_n = E_0 I\{N = n\} 1_B L_N.$$

Because $P_0(N < \infty) = P_1(N < \infty) = 1$,

$$\begin{aligned} P_1(B) &= P_1(B, N < \infty) = \sum_{n=1}^{\infty} P_1(N = n, B) \\ &= \sum_{n=1}^{\infty} E_0 I\{N = n\} 1_B L_N = E_0 I\{N < \infty\} 1_B L_N = E_0 1_B L_N. \quad \square \end{aligned}$$

If P_0 and P_1 are restricted and only considered as measures on $\mathcal{F}_N$, this result asserts that P_1 has density L_N with respect to P_0 . Theorem 5.4 follows from this. It can be shown that any σ -finite measure μ is equivalent to, or has the same null sets as, a probability measure. So in Theorem 5.4 we can

assume that the dominating measure μ is a probability measure. If the X_i are i.i.d. with density f_θ under P_θ , viewed as P_1 in Wald's fundamental identity, and they are i.i.d. from μ under P_0 , then the density for the restriction of P_θ to $\mathcal{F}_N$ is

$$L_N = \prod_{i=1}^N f(X_i),$$

with respect to the restriction of P_0 to $\mathcal{F}_N$.

20.3 Optimal Stopping

This section provides an introduction to the theory of optimal stopping, used to select the best stopping time N for a sequential experiment. The main ideas are developed in the context of Bayesian hypothesis testing.

Given $\Theta = \theta$, let potential observations $X_1, X_2, \dots$ be i.i.d. from Q_θ , and consider testing $H_0 : \Theta \in \Omega_0$ versus $H_1 : \Theta \in \Omega_1$. To be specific in our goals we proceed in a decision-theoretic fashion, assigning costs for the consequences of our inferential actions, with additional costs to perform the experiment and collect data. Inferential actions and stopping times are chosen to minimize expected costs.

After data collection, one of the hypotheses, H_0 or H_1 , will be accepted. Let variable A specify this action, with $A = 0$ if we accept H_0 and $A = 1$ if we accept H_1 . This action depends on the observed data $\mathcal{F}_N$, so A must be $\mathcal{F}_N$ measurable. Let $L(\Theta)$ denote the loss if we make the wrong decision: $A = 0$ when $\Theta \in \Omega_1$, or $A = 1$ when $\Theta \in \Omega_0$. The following result characterizes an optimal action A .

Theorem 20.7. *The inferential risk associated with action A , given by*

$$R(A) = EL(\Theta)[I\{A = 0, \Theta \in \Omega_1\} + I\{A = 1, \Theta \in \Omega_0\}],$$

will be minimal if

$$A = 0 \text{ on } E[L(\Theta)I\{\Theta \in \Omega_1\} \mid \mathcal{F}_N] < E[L(\Theta)I\{\Theta \in \Omega_0\} \mid \mathcal{F}_N]$$

and

$$A = 1 \text{ on } E[L(\Theta)I\{\Theta \in \Omega_1\} \mid \mathcal{F}_N] > E[L(\Theta)I\{\Theta \in \Omega_0\} \mid \mathcal{F}_N].$$

Proof. Because A is $\mathcal{F}_N$ measurable,

$$E[L(\Theta)I\{A = 0\}I\{\Theta \in \Omega_1\} \mid \mathcal{F}_N] = I\{A = 0\}E[L(\Theta)I\{\Theta \in \Omega_1\} \mid \mathcal{F}_N]$$

and

$$E[L(\Theta)I\{A = 1\}I\{\Theta \in \Omega_0\} \mid \mathcal{F}_N] = I\{A = 1\}E[L(\Theta)I\{\Theta \in \Omega_0\} \mid \mathcal{F}_N].$$

So by smoothing,

$$\begin{aligned}
 R(A) &= EE \left[L(\Theta) [I\{A = 0, \Theta \in \Omega_1\} + I\{A = 1, \Theta \in \Omega_0\}] \mid \mathcal{F}_N \right] \\
 &= E \left[I\{A = 0\} E[L(\Theta) I\{\Theta \in \Omega_1\} \mid \mathcal{F}_N] \right. \\
 &\quad \left. + I\{A = 1\} E[L(\Theta) I\{\Theta \in \Omega_0\} \mid \mathcal{F}_N] \right] \\
 &\geq E \min \left\{ E[L(\Theta) I\{\Theta \in \Omega_1\} \mid \mathcal{F}_N], E[L(\Theta) I\{\Theta \in \Omega_0\} \mid \mathcal{F}_N] \right\}.
 \end{aligned}$$

This bound is achieved if A has the form indicated in the theorem. □

Using this result, if we define

$$\rho_N = \min \left\{ E[L(\Theta) I\{\Theta \in \Omega_1\} \mid \mathcal{F}_N], E[L(\Theta) I\{\Theta \in \Omega_0\} \mid \mathcal{F}_N] \right\},$$

the inferential risk with an optimal action A is $E\rho_N$, and an optimal stopping rule N should balance this risk against expected costs running the experiment. A simple assumption, natural and fairly appropriate in many cases, is that each observation costs some fixed amount $c > 0$. The total cost to run the experiment is then cN , and an optimal stopping rule N minimizes

$$E[cN + \rho_N]. \tag{20.5}$$

To illustrate some ideas useful in a broader context, let us now restrict attention to a simple example, testing $H_0 : \Theta \leq 1/2$ versus $H_1 : \Theta > 1/2$ with Θ the success probability for a sequence of independent Bernoulli trials. We develop a recursive method, called backwards induction, to find an optimal stopping time. For the loss function, let us take

$$L(\Theta) = K|\Theta - 1/2|.$$

This function decreases as Θ tends to $1/2$, which seems natural because incorrect inference when Θ is near $1/2$ should be less of a concern than incorrect inference when Θ lies farther from $1/2$. Finally, for a prior distribution assume

$$\Theta \sim \text{Beta}(\alpha, \beta).$$

To begin, let us consider our inferential risk ρ_0 if we stop immediately with no data collection. Noting that

$$L(\Theta) I\{\Theta > 1/2\} = \frac{1}{2}K|\Theta - 1/2| + \frac{1}{4}K - \frac{1}{2}K(1 - \Theta)$$

and

$$L(\Theta) I\{\Theta \leq 1/2\} = \frac{1}{2}K|\Theta - 1/2| + \frac{1}{4}K - \frac{1}{2}K\Theta,$$

$$\rho_0 = \frac{1}{2}KE|\Theta - 1/2| + \frac{1}{4}K - \frac{1}{2}K \frac{\max\{\alpha, \beta\}}{\alpha + \beta}.$$

In form, ρ_0 is a function of α and β , $\rho_0 = H(\alpha, \beta)$, but the specific representation given is convenient. Calculations to find ρ_N are similar, but involve conditional expectations given $\mathcal{F}_N$. By Wald's identity (Theorem 20.6), the likelihood function for the data is proportional to

$$\theta^{S_N}(1 - \theta)^{N - S_N},$$

where $S_N = X_1 + \cdots + X_N$. Calculations identical to those in Example 7.3 show that

$$\Theta | \mathcal{F}_N \sim \text{Beta}(\alpha_N, \beta_N),$$

where

$$\alpha_n = \alpha + S_n \text{ and } \beta_n = \beta + n - S_n.$$

The calculations for ρ_0 involved expectations for functions of Θ , which can be viewed as integrals against the $\text{Beta}(\alpha, \beta)$ distributions. The calculations for ρ_N are identical, except that the expectations are integrals against $\text{Beta}(\alpha_N, \beta_N)$, the posterior distribution for Θ given $\mathcal{F}_N$. Using this observation,

$$\rho_N = \frac{1}{2}KE[|\Theta - 1/2| | \mathcal{F}_N] + \frac{1}{4}K - \frac{1}{2}K \frac{\max\{\alpha_N, \beta_N\}}{\alpha_N + \beta_N}.$$

By smoothing,

$$E|\Theta - 1/2| = EE[|\Theta - 1/2| | \mathcal{F}_N],$$

so according to criteria (20.5), our stopping time N should be chosen to minimize

$$\frac{1}{2}KE|\Theta - 1/2| + \frac{1}{4}K - \frac{1}{2}KE \left[\frac{\max\{\alpha_N, \beta_N\}}{\alpha_N + \beta_N} \right] + cEN.$$

Equivalently, since the first two terms here are independent of N , the stopping time N should be chosen to maximize

$$\frac{1}{2}KE \left[\frac{\max\{\alpha_N, \beta_N\}}{\alpha_N + \beta_N} \right] - cEN. \quad (20.6)$$

This expression has an interesting interpretation as the expected reward in a "guess the next observation" game. In this game, c is the cost for each observation, and after sampling, the player tries to guess whether X_{N+1} will be 0 or 1, winning $K/2$ for a correct guess.

Recursive algorithms to maximize (20.6) or minimize (20.5) rely on the conditional independence of the X_i given Θ . Since

$$P(X_{k+1} = x_1, \dots, X_{k+n} = x_n | \Theta, \mathcal{F}_k) = \Theta^{s_n}(1 - \Theta)^{n - s_n},$$

where $s_n = x_1 + \cdots + x_k$, smoothing gives

$$P(X_{k+1} = x_1, \dots, X_{k+n} = x_n | \mathcal{F}_k) = E[\Theta^{s_n}(1 - \Theta)^{n - s_n} | \mathcal{F}_k].$$

Since the variable of interest here is a function of Θ , this could be computed as an integral against $\text{Beta}(\alpha_k, \beta_k)$, the posterior distribution of Θ given $\mathcal{F}_k$. Formally, the answer would be exactly the same as $P(X_1 = x_1, \dots, X_n = x_n)$ if the prior distribution were $\text{Beta}(\alpha_k, \beta_k)$. So the joint distribution of future observations given $\mathcal{F}_k$ only depends on the posterior distribution of Θ . The observed data $X_1, \dots, X_k$, once they have been used to compute this posterior distribution, have no other effect on the distribution of future observations. The posterior distributions, characterized by the sequence (α_n, β_n) , $n \geq 0$, form a Markov chain (see Section 15.3), and given (α_k, β_k) , the initial observations $X_1, \dots, X_k$ and the future observations $X_{k+1}, X_{k+2}, \dots$ are independent.

To use the Markov structure described, let $V(\alpha, \beta)$ denote the supremum of (20.6),

$$V(\alpha, \beta) = \sup_{\text{stopping times } N} \left[\frac{1}{2}KE \left(\frac{\max\{\alpha_N, \beta_N\}}{\alpha_N + \beta_N} \right) - cEN \right],$$

called the *value* of the game. Suppose the player takes an initial observation and proceeds after this observation stopping optimally at some later stage. How much should he expect to win? Given X_1 , future observations will evolve as if the prior were $\text{Beta}(\alpha_1, \beta_1)$ and we had no data. Since we have to pay c for the first observation, the expected winnings will be

$$EV(\alpha_1, \beta_1) - c.$$

Since $\alpha_1 = \alpha + X_1$, $\beta_1 = \beta + 1 - X_1$, and

$$P(X_1 = 1) = EP(X_1 = 1|\Theta) = E\Theta = \frac{\alpha}{\alpha + \beta},$$

the expected winnings can be written as

$$\frac{\alpha V(\alpha + 1, \beta) + \beta V(\alpha, \beta + 1)}{\alpha + \beta} - c.$$

If instead the player stops immediately, he will win

$$\frac{1}{2}K \frac{\max\{\alpha, \beta\}}{\alpha + \beta}.$$

The optimal expectation must be the larger of these, so

$$V(\alpha, \beta) = \max \left\{ \frac{1}{2}K \frac{\max\{\alpha, \beta\}}{\alpha + \beta}, \frac{\alpha V(\alpha + 1, \beta) + \beta V(\alpha, \beta + 1)}{\alpha + \beta} - c \right\}. \quad (20.7)$$

This key equation can be used to calculate V recursively. To get started, if the sum $\alpha_n + \beta_n = \alpha + \beta + n$ is extremely large, the value of information from a single additional observation cannot offset its cost c , and so $V(\alpha_n, \beta_n)$ will be

$$V(\alpha_n, \beta_n) = \frac{1}{2}K \frac{\max\{\alpha_n, \beta_n\}}{\alpha + \beta + n}.$$

There are $n + 1$ possible values for (α_n, β_n) : $(\alpha, \beta + n)$, $(\alpha + 1, \beta + n - 1)$, $\dots$, $(\alpha + n, \beta)$ and the values for V on this grid can be saved as a vector or array on a computer. Using these values, V can be computed at $(\alpha, \beta + n - 1)$, $\dots$, $(\alpha + n - 1, \beta)$, all of the possible values for $(\alpha_{n-1}, \beta_{n-1})$, using (20.7). Continuing in this fashion we will eventually find V at all of the possible values for (α_k, β_k) , $k = 0, \dots, n$. This recursive approach to calculating V is called *backwards induction*.

Once V has been computed, it is easy to characterize an optimal stopping time N . From the discussion above, it should be clear that stopping immediately will be optimal if

$$\frac{1}{2}K \frac{\max\{\alpha, \beta\}}{\alpha + \beta} \geq \frac{\alpha V(\alpha + 1, \beta) + \beta V(\alpha, \beta + 1)}{\alpha + \beta} - c.$$

With the Markov structure, the decision to stop or continue at stage k will be the same as the decision to stop or continue initially if the prior were $\text{Beta}(\alpha_k, \beta_k)$. Thus it is optimal to stop the first time k that

$$\frac{1}{2}K \frac{\max\{\alpha_k, \beta_k\}}{\alpha + \beta + k} \geq \frac{\alpha_k V(\alpha_k + 1, \beta_k) + \beta_k V(\alpha_k, \beta_k + 1)}{\alpha + \beta + k} - c.$$

Given the data, the left-hand side here is what you expect to win if you stop at stage k , and the right-hand side is what you can expect to win if you take an additional observation and stop optimally at some later time.

If $\alpha = \beta = 1$, so the prior distribution for Θ is the uniform distribution on $(0, 1)$, and if $K = 200$ and $c = 1$, the optimal stopping time continues until (α_n, β_n) leaves the set

$$\{(1, 1), (2, 1), (1, 2), (2, 2), (3, 2), (2, 3), (3, 3), (4, 3), (3, 4), (4, 4), \\ (5, 4), (4, 5), (5, 5), (6, 5), (5, 6), (6, 6), (7, 6), (6, 7), (7, 7)\}.$$

The expected value for this stopping time is 3.12683, the chance of a correct guess for X_{N+1} is 0.71447, and the inferential risk is 3.55256. In contrast, the best fixed sample design will sample three observations, and the inferential risk with this sample size is 5.0. The sequential experiment is more efficient. Although the expected sample size is slightly larger, its inferential risk is 29% smaller.

20.4 Sequential Probability Ratio Test

The sequential probability ratio test is suggested in Wald (1947) for simple versus simple testing with i.i.d. observations with optional stopping. Take

$\Omega = \{0, 1\}$, let $X_1, X_2, \dots$ be i.i.d. from Q_θ with density q_θ , and consider testing $H_0 : \theta = 0$ versus $H_1 : \theta = 1$. Define

$$L_n = L_n(X_1, \dots, X_n) = \frac{\prod_{i=1}^n q_1(X_i)}{\prod_{i=1}^n q_0(X_i)},$$

the likelihood ratio for the first n observations. By convention, for $n = 0$ we take $L_0 = 1$. We know from the Neyman–Pearson theory in Section 12.2, that with a fixed sample size the best test rejects H_0 according to the size of L_n . The sequential probability ratio test (SPRT) has a similar feel. At each stage, the researcher has three options: stop and accept H_0 , stop and accept H_1 , or continue sampling. For the SPRT these options are resolved by comparing the likelihood ratio with two critical values $A < 1 < B$ in the following manner;

$$\begin{array}{ll} \text{if } L_n \in (A, B), & \text{take another observation;} \\ \text{if } L_n \geq B, & \text{reject } H_0; \\ \text{if } L_n \leq A, & \text{accept } H_0. \end{array} \quad (20.8)$$

Formally, the sample size³ for this SPRT is then

$$N = \inf \{n : L_n \notin (A, B)\}.$$

To understand the optimality properties of the SPRT, let us consider this testing problem from a Bayesian perspective. Let Θ be a Bernoulli variable with success probability $\pi = P(\Theta = 1)$, and given $\Theta = \theta$, potential observations $X_1, X_2, \dots$ be i.i.d. from, Q_θ . Because

$$P(\Theta = \theta | X_1, \dots, X_n) \propto_\theta P(\Theta = \theta) \prod_{i=1}^n q_\theta(X_i),$$

it is not hard to show that

$$\pi_n \stackrel{\text{def}}{=} P(\Theta = 1 | X_1, \dots, X_n) = \frac{\pi L_n}{1 - \pi + \pi L_n}, \quad (20.9)$$

an increasing function of L_n . For convenience, let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Given Θ the data are i.i.d., and so, for any Borel set $B \in \mathbb{R}^k$,

$$P[(X_{n+1}, \dots, X_{n+k}) \in B \mid \Theta, \mathcal{F}_n] = Q_\Theta^k(B).$$

So by smoothing,

$$\begin{aligned} P[(X_{n+1}, \dots, X_{n+k}) \in B \mid \mathcal{F}_n] &= E[Q_\Theta^n(B) | \mathcal{F}_n] \\ &= (1 - \pi_n) Q_0^k(B) + \pi_n Q_1^k(B). \end{aligned}$$

If data $X_1, \dots, X_n$ are observed, the conditional distribution of the remaining observations is a function only of π_n and does not depend in any other way

³ Some authors also consider a procedure that stops immediately ($N = 0$) a SPRT.

on either n or values for the first n observations. This Markov structure is essentially the same as that for the testing problem considered in Section 20.3, and also holds more generally. See Problem 20.5.

As in the last section, let $L(\theta)$ denote the cost for accepting the wrong hypothesis. If we stop without collecting any data, our minimal inferential risk, taking the smaller of the risks for the two actions, is

$$\rho(\pi) = \min\{\pi L(1), (1 - \pi)L(0)\}.$$

If we collect data, as in Theorem 7.1 it will be optimal to minimize posterior inferential risk. Since optional stopping does not change the likelihood, $P(\theta = 1 | \mathcal{F}_N) = \pi_N$, and the minimal posterior inferential risk, again minimizing over the two actions, is just $\rho(\pi_N)$.

To have a definite design objective, let us now assume, as in Section 20.3, that each observation costs an amount c , so the total sampling costs with a stopping rule N will be cN . Then the total risk for a stopping time N and $(\mathcal{F}_N$ measurable) test function φ will be

$$\begin{aligned} R(\pi, N, \varphi) &= E[cN + L(1)I\{\theta = 1\}(1 - \varphi) + L(0)I\{\theta = 0\}\varphi] \\ &= \pi cE_1N + (1 - \pi)cE_0N + \pi L(1)E_1(1 - \varphi) + (1 - \pi)L(0)E_0\varphi, \end{aligned}$$

where the second equality follows by conditioning on θ , with E_0 and E_1 denoting conditional expectation given $\theta = 0$ and $\theta = 1$, respectively. Note that $R(\pi, N, \varphi)$ is a linear function of π . By the argument above, if φ is chosen optimally the posterior inferential risk is $\rho(\pi_N)$, and so

$$R(\pi, N) \stackrel{\text{def}}{=} \inf_{\varphi} R(\pi, N, \varphi) = E[cN + \rho(\pi_N)].$$

Let $R(\pi)$ be the optimal risk, obtained minimizing over stopping times N :

$$R(\pi) = \inf_N R(\pi, N) = \inf_{N, \varphi} R(\pi, N, \varphi).$$

Note that since $R(\pi)$ is the infimum of a collection of linear functions, it must be concave.

Let us next consider whether stopping immediately is optimal. Since there are no sampling costs, the risk for the stopping time $N = 0$ is $\rho(\pi)$. The best possible risk, taking at least one observation, is

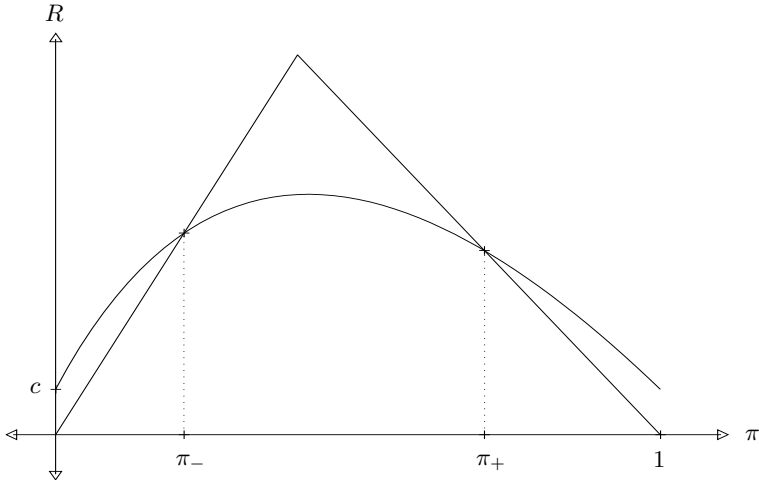
$$R_1(\pi) = \inf_{N \geq 1} R(\pi, N),$$

which is concave by the same reasoning as that for R . Comparing these risks, stopping immediately is optimal only if

$$\rho(\pi) \leq R_1(\pi).$$

The functions ρ and R_1 are graphed in [Figure 20.1](#), and the values $\pi_{\pm}$ where the two functions agree are indicated on the horizontal axis. As pictured, the function R_1 is continuous,⁴ and approaches c as $\pi \rightarrow 0$ or $\pi \rightarrow 1$. In some

⁴ Because R_1 is concave, continuity is immediate on $(0, 1)$. The argument for continuity at the endpoints 0 and 1 is more delicate.


 Fig. 20.1. ρ and R_1 .

cases, the function ρ may lie entirely below R_1 . Barring this possibility, if the functions ever cross, the values $\pi_- \leq \pi_+$ are uniquely determined. By the Markov structure, the decision to stop or continue at stage n should be formed in the same fashion after replacing the prior probability π with the posterior probability π_n . Thus the rule that stops the first time $\pi_n \notin (\pi_-, \pi_+)$ is optimal.

Theorem 20.8. *If $R_1(\pi) < \rho(\pi)$, so $N = 0$ is suboptimal, then the SPRT with*

$$A = \frac{1 - \pi}{\pi} \frac{\pi_-}{1 - \pi_-} \quad \text{and} \quad B = \frac{1 - \pi}{\pi} \frac{\pi_+}{1 - \pi_+}$$

is optimal, minimizing $R(\pi, N)$ over all stopping times N .

Proof. This follows from the discussion above and the monotonic relationship between π_n and L_n in (20.9). With A and B as defined in this theorem, straightforward algebra shows that $\pi_n \in (\pi_-, \pi_+)$ if and only if $L_n \in (A, B)$. $\square$

To allow comparisons with procedures that may accept or reject H_0 in a suboptimal fashion, let N and φ denote the sample size and test function for a sequential procedure. As usual, $E_\theta \varphi$ gives the chance of rejecting H_0 , so the error probabilities for this test are $\alpha_0 \stackrel{\text{def}}{=} E_0 \varphi$ and $\alpha_1 \stackrel{\text{def}}{=} 1 - E_1 \varphi$. The risk for this procedure in the Bayesian model is

$$R(\pi, N, \varphi) = \pi c E_1 N + (1 - \pi) c E_0 N + \pi L(1) \alpha_1 + (1 - \pi) L(0) \alpha_0,$$

a linear combination of α_0 , α_1 , $E_0 N$, and $E_1 N$. Because SPRTs are optimal, they must minimize the linear combination of the expected sample sizes $E_0 N$

and E_1N in this risk among all procedures with the same or better error probabilities. The following striking result of Wald and Wolfowitz (1948) is stronger, asserting that the SPRT simultaneously minimizes both of these expected sample sizes.

Theorem 20.9. *Let $\tilde{\alpha}_0$, $\tilde{\alpha}_1$, and $\tilde{N}$ be the error probabilities and sample size for a SPRT with $0 < \tilde{A} < 1 < \tilde{B} < \infty$. If α_0 and α_1 are error probabilities for a competing procedure (N, φ) , and if $\alpha_0 \leq \tilde{\alpha}_0$ and $\alpha_1 \leq \tilde{\alpha}_1$, then $E_0N \geq E_0\tilde{N}$ and $E_1N \geq E_1\tilde{N}$.*

The proof of this result is based on showing that the SPRT $\tilde{N}$ is optimal for Bayesian models with different values for the prior probability π , which may seem plausible because the loss structure for the problem depends on several values, c , $L(0)$, and $L(1)$, which can be varied, whereas the SPRT is completely specified by A and B .

A rescaling of costs just amounts to measuring them in different monetary units, and has no impact on a procedure's optimality. So let us assume that $L(0) + L(1) = 1$ and define $\omega = L(1)$, so $L(0) = 1 - \omega$. For notation, we write $\pi_{\pm} = \pi_{\pm}(c, \omega)$ and $R_1(\pi) = R_1(\pi, c, \omega)$ to indicate how these critical values and risk depend on ω and c .

Lemma 20.10. *For any values $0 < \hat{\pi}_- < \hat{\pi}_+ < 1$ there exist values $\omega \in (0, 1)$ and $c > 0$ such that $\pi_-(c, \omega) = \hat{\pi}_-$ and $\pi_+(c, \omega) = \hat{\pi}_+$.*

A careful proof of this lemma takes some work. It is not hard to argue that $R_1(\pi, c, \omega)$ is continuous, strictly increasing in c when π and ω are fixed, and tends to zero as $c \downarrow 0$, again with π and ω fixed. It follows that $\pi_{\pm}(c, \omega)$ are continuous, and that with ω fixed, $\pi_+(c, \omega)$ is an increasing function of c , $\pi_-(c, \omega)$ is a decreasing function of c , and $\pi_-(c, \omega) \rightarrow 0$ and $\pi_+(c, \omega) \rightarrow 1$ as $c \downarrow 0$. From this, for fixed ω , ratio A/B for the SPRT,

$$\frac{\pi_-(c, \omega)}{1 - \pi_-(c, \omega)} \frac{1 - \pi_+(c, \omega)}{\pi_+(c, \omega)},$$

is continuous in c and increases from 0 to 1 as c varies. The proof of the lemma is completed showing that if c is chosen to keep this ratio A/B fixed, then $\pi_+(c, \omega)$ will increase from 0 to 1 as ω varies over $(0, 1)$. Intuitively, this occurs because as ω increases, the risk for accepting H_0 increases while the risk for accepting H_0 decreases, leading to an increase in critical value $\pi_+(c, \omega)$ necessary to accept H_1 . A careful proof takes some care; details are available in Lehmann (1959).

Proof of Theorem 20.9. Given any value $\pi \in (0, 1)$, define $0 < \hat{\pi}_- < \pi < \hat{\pi}_+ < 1$ by

$$\hat{\pi}_- = \frac{\tilde{A}\pi}{\tilde{A}\pi + 1 - \pi} \quad \text{and} \quad \hat{\pi}_+ = \frac{\tilde{B}\pi}{\tilde{B}\pi + 1 - \pi}. \quad (20.10)$$

Using Lemma 20.10, choose ω and c so that

$$\pi_-(\omega, c) = \hat{\pi}_- \quad \text{and} \quad \pi_+(\omega, c) = \hat{\pi}_+. \quad (20.11)$$

By Theorem 20.8, with the loss structure (c, ω) and π as the prior probability, the optimal sequential procedure will be a SPRT with

$$A = \frac{1 - \pi}{\pi} \frac{\pi_-(\omega, c)}{1 - \pi_-(\omega, c)} \quad \text{and} \quad B = \frac{1 - \pi}{\pi} \frac{\pi_+(\omega, c)}{1 - \pi_+(\omega, c)}.$$

But by (20.10) and (20.11), $\tilde{A} = A$ and $\tilde{B} = B$, and so this SPRT is the same as the one with stopping time $\tilde{N}$ in the theorem. Because this SPRT minimizes risk,

$$\begin{aligned} \pi c E_1 \tilde{N} + (1 - \pi) c E_0 \tilde{N} + \pi L(1) \tilde{\alpha}_1 + (1 - \pi) L(0) \tilde{\alpha}_0 \\ \leq \pi c E_1 N + (1 - \pi) c E_0 N + \pi L(1) \alpha_1 + (1 - \pi) L(0) \alpha_0, \end{aligned}$$

which implies

$$\pi c E_1 \tilde{N} + (1 - \pi) c E_0 \tilde{N} \leq \pi c E_1 N + (1 - \pi) c E_0 N.$$

But $\pi \in (0, 1)$ is arbitrary, and this bound can hold for all $\pi \in (0, 1)$ only if $E_0 N \geq E_0 \tilde{N}$ and $E_0 N \geq E_0 \tilde{N}$. $\square$

20.5 Sequential Design

In other sections of this chapter, the decision concerning when to stop experimentation has been the central design issue. Here we move beyond stopping problems and consider procedures with other design options. The first example, called *stochastic approximation*, concerns adaptive variable selection in regression models. In the other example we consider allocation, or bandit, problems.

Stochastic Approximation

Let Q_x , $x \in \mathbb{R}$, be the distribution for a response variable Y when an input variable X , chosen by the researcher, equals x . The mean of Q_x ,

$$f(x) \stackrel{\text{def}}{=} E[Y|X = x] = \int y dQ_x(y),$$

is called the *regression function*. Let $\sigma^2(x)$ denote the variance of Q_x ,

$$\sigma^2(x) = \int (y - f(x))^2 dQ_x(y).$$

Stochastic approximation, introduced in Robbins and Monroe (1951), considers situations in which the input variables are chosen adaptively. Specifically, X_1 is a constant, and for $n \geq 1$, X_{n+1} is a function of the first n

observations. So if $\mathcal{F}_n$ is the σ -field representing information from the first n observations,

$$\mathcal{F}_n = \sigma(X_1, Y_1, \dots, X_n, Y_n),$$

X_{n+1} is $\mathcal{F}_n$ measurable. Conditional distributions for the Y_n are given by

$$Y_n | \mathcal{F}_{n-1} \sim Q_{X_n}.$$

If the mechanism to select independent variables is specified, these conditional distributions determine the joint distribution for the data.

Let t be a fixed constant representing a target value for the regression function, and let θ denote the value for the independent variable that achieves this target, so

$$f(\theta) = t.$$

The design objective in stochastic approximation is to find an adaptive strategy that drives the independent variables X_n to θ as quickly as possible. It should be clear that this will be possible only with a sequential approach.

Example 20.11. Bioassay experiments are designed to investigate the relationship between a dose level x and the chance of some response. The median⁵ effective dose, ED50, is defined as the dose that gives a 50% chance of response. If the variable Y is a response indicator and its conditional distribution given $X = x$ is a Bernoulli distribution with success probability $f(x)$, then ED50 will be θ if the target t is $1/2$.

If the regression function f is assumed to be increasing, then a natural strategy would be to decrease the independent variable X if the response were to lie above the target and increase X if the response were to lie below the target. A specific recursion suggested in Robbins and Monro (1951) takes

$$X_{n+1} = X_n - a_n(Y_n - t), \quad n \geq 1, \quad (20.12)$$

where a_n , $n \geq 1$, is a sequence of positive constants.

Theorem 20.12. *If f is continuous and strictly increasing, $\sum a_n^2 < \infty$, $\sum a_n = \infty$, and*

$$\sup_{x \in \mathbb{R}} \frac{\sigma(x) + |f(x)|}{1 + |x|} < \infty,$$

then $X_n \rightarrow \theta$ almost surely as $n \rightarrow \infty$.

Asymptotic normality for X_n can be established under slightly stronger assumptions using results of Fabian (1968). In particular, $a_n = c/n$ gives the best rate of convergence. With this choice and suitable regularity,

⁵ The language here, though somewhat natural, is a bit unfortunate because this dose is not in any natural sense the median for any list or distribution for doses.

$$\sqrt{n}(X_n - \theta) \Rightarrow N\left(0, \frac{c^2 \sigma^2(\theta)}{2cf'(\theta) - 1}\right),$$

provided $2c > 1/f'(\theta)$. The asymptotic variance here is minimized taking $c = 1/f'(\theta)$. For more discussion, see the review articles by Ruppert (1991) and Lai (2003).

In practice, procedures seeking the maximum of a regression function may have even more practical value. Kiefer and Wolfowitz (1952) suggest an approach similar to that of Robbins and Monro. Let b_n , $n \geq 1$ be positive constants decreasing to zero. Responses are observed in pairs with the independent variable set to $X_n \pm b_n$. Then the conditional mean of $Z_n = (Y_{n+} - Y_{n-})/(2c_n)$ will be approximately $f'(X_n)$, and their recursion takes

$$X_{n+1} = X_n - a_n Z_n, \quad n \geq 1,$$

again with a_n , $n \geq 1$, a sequence of constants.

In industrial applications, the independent variables X_n are often multivariate, and the response surface methodology suggested in Box and Wilson (1951) has been popular. Experimentation proceeds in stages. For early stages, the independent variables are selected to allow a linear fit to estimate the gradient of the regression function. At successive stages, this information about the gradient is used to shift to a region with higher average response. Then at later stages richer designs are used that allow quadratic models to be fit. The maximum of the fitted quadratic is then taken as the estimate for the maximum of the regression function.

Bandit Problems

Bandit or allocation problems have a rich history and literature. We only touch on a few ideas here to try to give a feel for the general area. The “bandit” language refers to a slot machine for gambling, operated by pulling a lever (arm) and called informally a *one-armed bandit*. Playing the machine repeatedly, a gambler will receive a sequence of rewards, until he tires or runs out of quarters. If the gaming establishment has several machines, then the gambler may choose to switch among them, playing different arms over time. Our main concern is to find an optimal strategy for the gambler, that identifies the best arm to play at each stage.

Mathematically, bandit problems can be formulated in various ways. A special case of interest in statistics is discussed here, although extensions are mentioned. For more extensive developments, see Berry and Fristedt (1985).

At each stage $n \geq 0$, the researcher chooses among k arms. In a clinical setting, “arms” might correspond to giving one of k treatments for some medical condition. Each time an arm is played, the researcher observes a random variable (or vector) X , and the distribution of this variable for arm a is governed by an unknown random parameter θ_a . These parameters $\theta_1, \dots, \theta_k$ are independent with prior marginal distributions $\theta_a \sim \pi_0(a)$, $a = 1, \dots, k$.

Let $\mathcal{F}_n = \sigma(X_0, \dots, X_{n-1})$, the information from the first n observations, and let A_n denote the arm played at stage $n \geq 0$. Since A_n must be chosen based on past data, A_n is $\mathcal{F}_n$ measurable. Distributions for the observations, given the arm played and the value for the parameter of the arm, are denoted $Q(a, \theta_a)$. If a strategy to select arms to play has been fixed, then the joint distribution for the observations X_n , $n \geq 1$, is determined by the conditional distributions

$$X_n | \mathcal{F}_n, \Theta \sim Q(A_n, \Theta_{A_n}).$$

Every time arm a is played, the researcher receives a reward $r(a, \Theta_a)$, but the value of this reward if it is acquired at stage n is discounted geometrically by the factor β^n for some constant $\beta \in (0, 1)$, called the *discount factor*. For regularity, we assume the reward function $r(\cdot, \cdot)$ is bounded. The total discounted reward, if arms are played indefinitely, is

$$\sum_{n=0}^{\infty} r(A_n, \Theta_{A_n}) \beta^n,$$

and the design objective is to maximize the expected value of this variable. Using a conditioning argument the expectation V of this variable can be expressed in another way. If π is an arbitrary probability measure, define

$$r(a, \pi) = \int r(a, \theta) d\pi(\theta),$$

and let $\pi_n(a)$ denote the conditional distribution for Θ_a given $\mathcal{F}_n$. Then

$$E[r(A_n, \Theta_{A_n}) | \mathcal{F}_n] = r(A_n, \pi_n(A_n)).$$

Using Fubini's theorem to justify interchanging expectation and summation,

$$\begin{aligned} V &= \sum_{n=0}^{\infty} \beta^n E r(A_n, \Theta_{A_n}) = \sum_{n=0}^{\infty} \beta^n E E[r(A_n, \Theta_{A_n}) | \mathcal{F}_n] \\ &= E \sum_{n=0}^{\infty} \beta^n r(A_n, \pi_n(A_n)). \end{aligned} \quad (20.13)$$

In this form, the Markov structure of this allocation problem is clearer. When arm a is played, the posterior distributions $\pi_n(a)$, $n \geq 0$, evolve as a time homogeneous Markov chain, the same structure noted in Sections 20.3 and 20.4, and this is really the intrinsic structure needed for Theorem 20.13 below. So a Markov formulation for bandit allocation problems, seen in much of the literature, is formally more general, and also makes application to scheduling problems in operations research more evident.

An *allocation index* ν for arm a is a function of $\pi_n(a)$ (the current state of arm a), determined solely by the reward function $r(a, \cdot)$ for arm a and by the family $Q(a, \cdot)$ that implicitly determines the stochastic transition kernel

that dictates how posterior distributions for Θ_a evolve when arm a is played. An *index strategy* is an allocation procedure that always selects the arm with the largest allocation index.

A formula for allocation indices can be derived by considering simple two-armed bandit problems in which the researcher plays either arm a or an arm with a fixed reward λ . It is natural to calibrate indices so that the allocation index for the fixed arm is simply λ . These special bandit problems are actually stopping problems, for if it is ever correct to play the arm with a fixed reward, playing it at the current and all future stages is optimal. So we can restrict attention to strategies that play arm a before some stopping time τ , and the maximal expected discounted reward is

$$H(a, \pi_0(a), \lambda) = \sup_{\tau} E \left[\sum_{n=0}^{\tau-1} \beta^n r(a, \pi_n(a)) + \lambda \frac{\beta^{\tau}}{1-\beta} \right].$$

If λ is large enough, stopping immediately ($\tau = 0$) is optimal, and we have

$$H(a, \pi_0(a), \lambda) = \frac{\lambda}{1-\beta}.$$

But for sufficiently small λ , $\tau = 0$ is suboptimal, and arm a should be played at least once. In this case,

$$H(a, \pi_0(a), \lambda) > \frac{\lambda}{1-\beta}.$$

If index strategies are optimal in these problems, since the index for the fixed arm is λ , the initial index for arm a will be at most λ in the former case, and will exceed λ in the latter. So the index for arm a will be the smallest value λ where stopping immediately is optimal, the critical value dividing these two regions. Thus

$$\nu(a, \pi_0(a)) = \inf \left\{ \lambda : (1-\beta)H(a, \pi_0(a), \lambda) = \lambda \right\} \quad (20.14)$$

should be the initial allocation index for arm a . Allocation indices at later stages are obtained in the same way, replacing $\pi_0(a)$ with posterior distribution $\pi_n(a)$, in effect treating the posterior distribution as the prior distribution in the stopping problems.

Theorem 20.13 (Gittins). *An index strategy, with indices given by (20.14), is optimal for the k -armed bandit problem, maximizing the expected discounted reward V in (20.13).*

This beautiful result first appears in Gittins and Jones (1974). Whittle (1980) gives an elegant proof. Gittins (1979) gives a characterization of allocation indices using a notion of *forwards induction*, and Katehakis and Veinott (1987) relates the index to the value for a game based on a single arm, with the option of “restarting” at any stage.

20.6 Problems

1. *Two-stage procedures.* Let $X_1, X_2, \dots$ be i.i.d. from $N(\mu, \sigma^2)$. In a two-stage sequential procedure, the sample size N_1 for the first stage is a fixed constant, and the sample size N_2 for the second stage is based on observations $X_1, \dots, X_{N_1}$ from the first stage. Let $N = N_1 + N_2$ denote the total number of observations, let $\bar{X}_1$ and $\bar{X}_2$ denote sample averages for the first and second stages,

$$\bar{X}_1 = \frac{X_1 + \dots + X_{N_1}}{N_1} \quad \text{and} \quad \bar{X}_2 = \frac{X_{N_1+1} + \dots + X_N}{N_2},$$

and let $\bar{X}$ denote the average of all N observations. For this problem, assume that N_2 is a function of

$$S_1^2 = \frac{1}{N_1 - 1} \sum_{i=1}^{N_1} (X_i - \bar{X}_1)^2,$$

the sample variance for the first stage. For convenience, assume $N_2 \geq 1$ almost surely.

- Use smoothing to find the distribution of $\sqrt{N_2}(\bar{X}_2 - \mu)$.
 - Show that $\sqrt{N_2}(\bar{X}_2 - \mu)$ and $(\bar{X}_1, S_1)$ are independent.
 - Show that S_1 and $\sqrt{N}(\bar{X} - \mu)$ are independent.
 - Determine the distribution of $T = \sqrt{N}(\bar{X} - \mu)/S_1$ and give a confidence interval for μ with coverage probability (exactly) $1 - \alpha$.
 - The second-stage sample size N_2 is a function of S_1 . Suggest a choice for this function if the researcher would like a confidence interval for μ with width at most some fixed value w .
- If N_1 and N_2 are stopping times with respect to the same filtration $\mathcal{F}_n$, $n \geq 0$, show that $N_1 \wedge N_2$ and $N_1 \vee N_2$ are also stopping times.
 - Extend Theorem 20.6, giving an identity for $P_1(B)$, $B \in \mathcal{F}_N$, that holds if $P_1(N < \infty) = 1$ but $P_0(N = \infty) > 0$.
 - Given $\Theta_1 = \theta_1$ and $\Theta_2 = \theta_2$, let $X_1, X_2, \dots$ and $Y_1, Y_2, \dots$ be independent with the X_i from a Bernoulli distribution with success probability θ_1 and the Y_i from a Bernoulli distribution with success probability θ_2 . Consider a sequential experiment in which pairs of these variables, $(X_1, Y_1), (X_2, Y_2), \dots$, are observed until a stopping time N . Assume that Θ_1 and Θ_2 are *a priori* independent, each uniformly distributed on $(0, 1)$. Consider testing $H_0 : \Theta_1 \leq \Theta_2$ versus $H_1 : \Theta_1 > \Theta_2$ with loss $K|\Theta_1 - \Theta_2|$ if we accept the wrong hypothesis and total sampling costs cN . Derive a recursive “backwards induction” algorithm to find the optimal stopping time N . This algorithm will involve the posterior inferential risk ρ_n .
 - Consider a Bayesian model in which $\Theta \sim \Lambda$ and given $\Theta = \theta$, X_i , $i \geq 1$, are i.i.d. from Q_θ . Let Λ_n denote the posterior distribution for Θ given $X_1, \dots, X_n$. Use smoothing to show that

$$P[(X_n + 1, \dots, X_{n+k}) \in B \mid X_1, \dots, X_n] = \int Q_\theta^n(B) d\Lambda_n(\theta).$$

This equation shows that the distribution of future observations given the past depends on the conditioning variables only through the posterior distribution Λ_n .

6. *Secretary problems.* Let $X_1, \dots, X_n$ be i.i.d. from a uniform distribution, let $B_{ij} = \{X_i < X_j\}$, and define

$$\mathcal{F}_k = \sigma(B_{ij} : 1 \leq i < j \leq k)$$

($\mathcal{F}_1 = \{\emptyset, \mathcal{E}\}$, the trivial σ -field), so that $\mathcal{F}_k$ provides information about the relative ranks of X_i , $i \leq k$, but not their values. Also, let

$$\mathcal{G}_k = \sigma(X_1, \dots, X_k),$$

so the filtration $\mathcal{G}$ has information about the actual values of the X_i . Let

$$p(N) = P(X_N > X_i, 1 \leq i \leq n, i \neq N),$$

the chance X_N is maximal.

- a) Take $n = 5$. Find a stopping time N with respect to $\mathcal{F}$ that maximizes $p(N)$.
 - b) Take $n = 2$. Find a stopping time N with respect to $\mathcal{G}$ that maximizes $p(N)$.
7. *Wald's identity.* Let $X_1, X_2, \dots$ be i.i.d. with $E|X_i| < \infty$ and mean $\mu = EX_i$. Define $S_n = X_1 + \dots + X_n$ and let N be a (positive) stopping time with respect to the filtration $\mathcal{F}$ generated by these variables,

$$\mathcal{F}_n = \sigma(X_1, \dots, X_n), \quad n \geq 1.$$

Assume $EN < \infty$. Wald's identity asserts that

$$ES_N = \mu EN.$$

- a) Use indicator variables and Fubini's theorem to show that

$$EN = \sum_{n=1}^{\infty} P(N \geq n).$$

- b) Prove Wald's identity if the X_i are nonnegative, $X_i \geq 0$. Hint: Show that

$$S_N = \sum_{n=1}^{\infty} X_n I\{N \geq n\},$$

and use Fubini's theorem. Independence, related to the condition that N is a stopping time, will play an important role.

- c) Prove Wald's identity if the stopping time N is bounded, $N \leq k$, almost surely for some $k \geq 1$. An argument like that for part (a) should suffice.
- d) Prove Wald's identity in general. Hint: Take

$$N_k = \min\{N, k\},$$

so that $N_k \uparrow N$ as $k \rightarrow \infty$. Using part (b), it should be enough to show that $E[S_N - S_{N_k}]$ tends to zero. But this expectation is $E(S_N - S_k)I\{N > k\}$ (explain why). Use dominated convergence and independence to show this expectation tends to zero.

8. *Power one tests and one-sided SPRTs.* Power one tests arise in a sequential setting if we reject H_0 whenever our sample size N is finite and we stop. Since $N = \infty$ is desirable when H_0 is correct, sampling costs in this case should be zero. For the simple versus simple model considered for the SPRT in Section 20.4, if c represents the cost per observation when $\Theta = 1$ and L the loss for stopping if $\Theta = 0$, our stopping time N should be chosen to minimize

$$E[LI\{N < \infty, \Theta = 0\} + cNI\{\Theta = 1\}] = (1 - \pi)LP_0(N < \infty) + c\pi E_1 N.$$

- a) Use a convexity argument to show that an optimal stopping will have form

$$N = \inf\{n \geq 0 : \pi_n > \pi_+\},$$

for some constant $\pi_+ \in (0, 1)$.

- b) Define $Y_n = \log[q_1(X_i)/q_0(X_i)]$ and take $S_n = Y_1 + \cdots + Y_n = \log L_n$. Introduce ladder times

$$\mathcal{T}_+ \stackrel{\text{def}}{=} \inf\{n \geq 1 : S_n > 0\}, \quad \mathcal{T}_- \stackrel{\text{def}}{=} \inf\{n \geq 1 : S_n \leq 0\}.$$

Note that if π_0 equals π_+ , then $\mathcal{T}_+$ will be the optimal stopping time in part (a). But $N = 0$ should also be optimal. Use this observation and the duality formula

$$P_0(\mathcal{T}_+ = \infty) = \frac{1}{E_0 \mathcal{T}_-}$$

to derive an explicit formula relating $\pi_+/(1 - \pi_+)$ to moments of the ladder variables $\mathcal{T}_\pm$.

9. Consider the mean square performance of the Robbins–Monro recursion with $t = 0$ and $a_n = 1/n$, so that

$$X_{n+1} = X_n - \frac{Y_n}{n}, \quad n \geq 1.$$

Assume that $f(x) = x$, so $\theta = 0$, and that the variance function is constant, $\sigma^2(x) = \sigma^2$.

- a) Let $\mu_n = EX_n$. Use a conditioning argument to derive a recursion relating μ_{n+1} to μ_n . Solve this recursion, expressing μ_n , $n \geq 2$, as a function of the initial value $\mu_1 = X_1$.
- b) Define $m_n = EX_n^2$, the mean square error at stage n . Use conditioning to derive a recursion relating m_{n+1} to m_n . Solve this recursion and show that $m_n \rightarrow 0$ as $n \rightarrow \infty$. Hint: Let $w_n = (n-1)^2 m_n$, $n \geq 1$. The recursion for w_n should be easy to solve.